07/01/2026

Code-B

Aakash
Medical | IIT-JEE | Foundations

Corporate Office : AESL, 3rd Floor, Incuspaze Campus-2, Plot No. 13, Sector-18, Udyog Vihar, Gurugram, Haryana - 122015, ***Ph.***+91-1244168300

MM : 720 NCERT Booster Test Series for NEET-2026_RM(P2)_NBTS-03B **Time : 180 Min.**

PHYSICS

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CHEMISTRY

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ZOOLOGY

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Hints and Solutions

PHYSICS

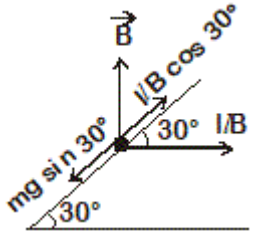
(1) **Answer :** (2)
Solution:
$X_L = \omega L = 2\pi \times 50 \times \frac{2}{\pi} = 200\Omega$
$X_C = \frac{1}{\omega C} = \frac{1}{100\pi \times \frac{10^{-4}}{\pi}} = 100\Omega$
$X = X_L - X_C = 200 - 100 = 100\ \Omega$
$R = 100\ \Omega$
$|Z| = \sqrt{x^2 + R^2} = 100\sqrt{2}$
$I = \frac{V}{Z} = \frac{220}{100\sqrt{2}} = \frac{11\sqrt{2}}{10}\text{A}$

(2) **Answer :** (2)
Hint:
$\vec{E} = -\frac{\partial V}{\partial x}\hat{i} - \frac{\partial V}{\partial y}\hat{j}$
Solution:
$\vec{E} = -2ky\hat{i} - 2kx\hat{j}$
$\left[\vec{E}\right] = -2k\hat{i} - 2k\hat{j}$
$\left|\vec{E}\right| = 2\sqrt{2}\,k$

(3) **Answer :** (3)
Solution:
$\varepsilon = -L\frac{di}{dt}$
$\therefore$ emf $\propto$ – (slope of i - t graph)

(4) **Answer :** (2)
Hint:
$R = \frac{ml}{ne^2A\tau}$
$R \propto \frac{1}{n}$ when L = constant, A = constant, τ = constant
$\frac{R_A}{R_B} = \frac{n_B}{n_A} = \frac{3}{1}$

(5) **Answer :** (3)
Solution:



For equilibrium,
$mg\sin 30^\circ = Bil\ \cos 30^\circ$
$i = \frac{mg}{lB}\tan 30^\circ$
$= \frac{0.5 \times 9.8}{0.25 \times \sqrt{3}} = 11.32\ \text{A}$

(6) **Answer :** (2)
Solution:
$100 = ni$
$i = \frac{100 \times 1}{50} = 2\ \text{A}$

(7) **Answer :** (1)
Solution:

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$H_{dc} = 2^2 Rt$ and $H_{ac} = i_{\text{rms}}^2 Rt = \left(\frac{2}{\sqrt{2}}\right)^2 Rt$

$\frac{H_{dc}}{H_{dc}} = 2$

(8) **Answer :** (4)
Solution:
E = 200 sin (100πt) cos(100πt)
E = 100 sin (200πt)
(∵ 2 sinθ cosθ = sin2θ)
⇒ E_0 = 100 V ; ω = 200π = 2πf
⇒ f = 100 Hz

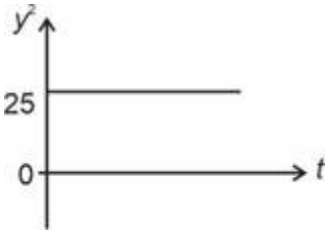
(9) **Answer :** (1)
Solution:
As potential drop across 2 μF will increase on insertion of dielectric, so charge on 2 μF capacitor will increase.

(10) **Answer :** (2)
Solution:
Moving from A to B
$V_B - V_A = [5 \times 10^{-3}(-10^3) + 15 + 1 \times 5] = 15$ volt

(11) **Answer :** (4)
Solution:
According to Gauss's law for magnetism, the net magnetic flux through any closed surface is always zero because magnetic filed lines form closed loops, and total magnetic flux linked with closed surface is always zero.

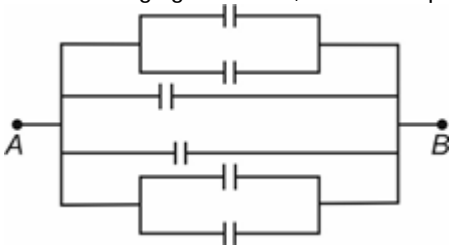
(12) **Answer :** (2)
Solution:
Bar magnet creates non-uniform magnetic field. Net force on bar magnet placed in uniform magnetic field is zero because the equal and opposite forces acting on its north and south poles cancel each other out..

(13) **Answer :** (1)
Hint:
$y_{\text{rms}} = \sqrt{<y^2>}$
Solution:



$y_{\text{rms}} = \sqrt{<y^2>} = \sqrt{25}$ = 5

(14) **Answer :** (2)
Hint:
Use $C_P = C_1 + C_2 + \ldots$
Solution:
After rearranging the circuit, effective capacitance can be given as



C = 6C
= 6 × 2 μF
= 12 μF

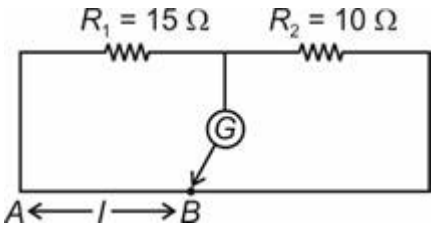
(15) **Answer :** (3)
Solution:
$\mu_r = 1 + \chi$, the magnetic susceptibility of diamagnetic material is slightly negative. The susceptibility of paramagnetic material is slightly positive and susceptibility of ferromagnetic material is large and positive.

(16) **Answer :** (4)
Solution:

With the rise of temperature conductivity of conductors (metals) decreases and drift speed decreases because relaxation time (τ) decreases.

$\left(v_d = \frac{eE\tau}{m}\right)$

(17) Answer : (3)
Solution:
Hint: When galvanometer shows no deflection, it is balance bridge condition.
Sol.: Let *AB* = *l* cm
BC = (100 – *l*) cm
At zero deflection of galvanometer



$\frac{R_1}{R_2} = \frac{R_{AB}}{R_{BC}} = \frac{l}{100-l}$

$\frac{15}{10} = \frac{l}{100-l}$

$\frac{3}{2} = \frac{l}{100-l}$

∴ 300 – 3*l* = 2*l*
5*l* = 300
l = 60 cm

(18) Answer : (3)
Solution:
For stable equilibrium,
P.E. is *U* = –*MB* cosθ

$= -\left(0.4\frac{\text{J}}{\text{T}}\right)\left(0.24\right) = -0.096\,\text{J}$

(19) Answer : (1)
Solution:

∵ $\phi = 5t^2$

∴ $\frac{d\phi}{dt} = 10t$

induced emf at *t* = 1 s will be

E_{in} = 10 × 1 = 10 V

∴ induced current $i = \frac{E_{in}}{R}$

$= \frac{10}{20}$

= 0.5 A

(20) Answer : (3)
Solution:

$\phi = \vec{B} \cdot \vec{A}, \quad \Rightarrow \phi_{\max}, \text{ when } \cos\theta = 1 \Rightarrow \theta = 0^\circ$

(21) Answer : (2)
Hint:

Common potential $V_{\text{common}} = \frac{C_1V_1 + C_2V_2}{C_1 + C_2}$

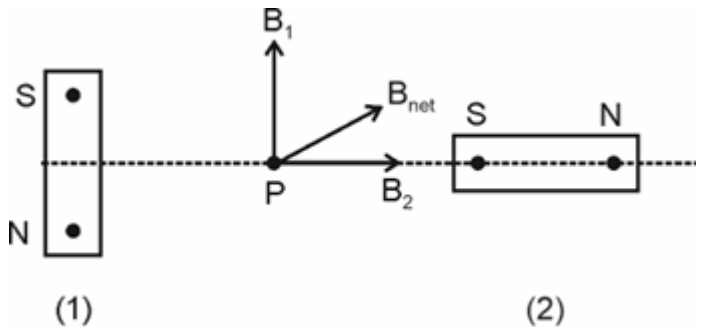
Solution:

$V = \frac{C_1V_1 + C_2V_2}{C_1 + C_2}$

$= \frac{20\times100 + 20\times0}{40} = 50\,\text{V}$

(22) Answer : (2)
Solution:

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(23) **Answer :** (4)
Hint:
$I_{\text{rms}} = \frac{I_{\text{peak}}}{\sqrt{2}}$ and $V_{\text{rms}} = \frac{V_{\text{peak}}}{\sqrt{2}}$
Solution:
Statement I is wrong because V_{rms} = 220 V in domestic AC circuit and peak voltage = $220\sqrt{2}$ V.
There could be a half time period in AC current which may have zero average value.

(24) **Answer :** (4)
Hint:
$I_s = \frac{\varphi}{I} = \frac{NAB}{K}$
Solution:
Current sensitivity of the galvanometer is defined as the deflection per unit current.

(25) **Answer :** (2)
Hint:
Voltage remains same and capacitance increases.
Solution:
Energy = $\frac{1}{2}KCV^2$,
$C' = KC$
$QV = KCV^2$,
$\frac{Q}{V} = KC$

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(26) **Answer :** (1)
Solution:
$r = \frac{\sqrt{2mk}}{qB}$
$r_p \propto \frac{\sqrt{m_p}}{e}$ and $r_\alpha \propto \frac{\sqrt{4m_p}}{2e} \Rightarrow r_\alpha \propto \frac{\sqrt{m_p}}{e}$
$T = \frac{2\pi r}{v} = \frac{2\pi m}{qB} \Rightarrow T \propto \frac{m}{q}$

(27) **Answer :** (3)
Solution:



$\frac{\mu_0 i}{2\pi(6-x)} = \frac{\mu_0(2i)}{2\pi x}$
$x = 12 - 2x$
$x = 4$ cm

(28) **Answer :** (4)
Solution:
Capacitive reactance is $X_C = \frac{1}{2\pi fC}$, so capacitive reactance decreases with increase in frequency of AC source.

(29) **Answer :** (2)
Solution:
$P = \frac{20}{\sqrt{2}} \times \frac{10}{\sqrt{2}} \times \cos\left(\frac{\pi}{3}\right)$ = 50 W

(30) **Answer :** (1)
Solution:
$J = \sigma E = \frac{E}{\rho}$
$\Rightarrow E = \rho J$

$20 = 4 \times 10^4 \times \rho$

$\rho = \frac{20}{4\times10^4}$

$= 5 \times 10^{-4}\ \Omega$

(31) **Answer :** (3)
Hint:
Lines of induced electric field are closed curve.
Solution:
This field is produced by time varying magnetic field.
This field is non-conservative in nature.

(32) **Answer :** (3)
Solution:

$T = 2\pi\sqrt{\frac{I}{MB}}$

$T = 2\pi\sqrt{\frac{9.8\times10^{-6}}{1280\pi^2\times10^{-5}\times0.049}}$

$T = 2\sqrt{\frac{9.8}{1280\times0.49}}$

$= 2\sqrt{\frac{20}{1280}} = \frac{1}{4}\ \text{sec}$

(33) **Answer :** (3)
Hint:
Work done by external force = Change in potential energy
Solution:
$U = -PE\cos\theta$
$U_i = -PE$
$U_f = -PE\cos\theta$
$W = -PE\cos\theta + PE$
$\frac{3PE}{2} = PE(1-\cos\theta)$
$\cos\theta = \frac{-1}{2}$
$\theta = 120°$

(34) **Answer :** (2)
Solution:
EMF induced between ends B and C is $V_C - V_A = Blv\sin60°$.
EMF induced between ends D and B is $V_A - V_D = Blv\sin60°$.
So, $V_C - V_D = 2\times\frac{\sqrt{3}}{2}Blv = \sqrt{3}Blv$.

(35) **Answer :** (2)
Solution:
$B = \mu_0 ni \Rightarrow 2\pi\times10^{-3} = 4\pi\times10^{-7}\times500\times2\times i \Rightarrow i = 5$ A

(36) **Answer :** (1)
Solution:
As, $R = \frac{\rho L}{A}$
$= \frac{\rho L^2 d}{m}$ ($d \rightarrow$density)
$\therefore R \propto L^2$

(37) **Answer :** (4)
Solution:
$L_{eq} = \frac{2L}{3} = \frac{2}{3}\times6 = 4\text{H}$

(38) **Answer :** (2)
Solution:
$P = \vec{F}\cdot\vec{v}$ and $\vec{F}_m \perp \vec{v}$

(39) **Answer :** (2)
Solution:
$E = -\frac{d\phi}{dt}$
$\phi = R\int idt = 10\times\frac{1}{2}\times5\times0.1$
= 2.5 weber

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(40) **Answer :** (2)
Solution:
$P = \frac{V^2}{R}$
$100 = \frac{400}{R+2}$
R = 2 Ω

(41) **Answer :** (1)
Solution:
$\vec{\tau} = \vec{M} \times \vec{B}$

$$\vec{\tau} = \begin{vmatrix} \hat{i} & \hat{j} & \hat{k} \\ 50 & 0 & 0 \\ 0.2 & 0.3 & 0 \end{vmatrix}$$

$\vec{\tau} = k\left[50 \times 0.3 - 0.2 \times 0\right] = \left(15\hat{k}\right)$ N m

(42) **Answer :** (4)
Solution:
Electric field inside conductor is zero only under electrostatic condition.

(43) **Answer :** (4)
Hint:
Force on a current element due to magnetic field, $d\vec{F} = i\left(\vec{dl} \times \vec{B}\right)$
Solution:
Effective length vector of the wire is $2b\hat{i}$
$\therefore \vec{F} = i\left(2b\hat{i} \times B_0\hat{i}\right) = 0$

(44) **Answer :** (1)
Solution:
Potential difference across terminal may be less or more than emf.

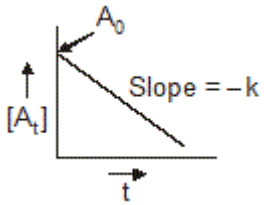
(45) **Answer :** (2)
Solution:
E = 0.5 (r + 12)
E = 0.25(r + 25)
$1 = \frac{2(r+12)}{(r+25)}$
r = 1 Ω

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CHEMISTRY

(46) **Answer :** (1)
Hint:
Catalyst increases reaction rates by reducing activation energies of both forward and backward reactions.
Solution:
A catalyst does not alter ΔG or K_{eq} of a reversible reaction.

(47) **Answer :** (3)
Hint:
For zero order reaction.
$[A_t] = [A_0] - kt$
Solution:



(48) **Answer :** (3)

Hint:

$t = \frac{2.303}{k}\log\frac{[A]_0}{[A]_t}$

Solution:

$t = \frac{2.303}{k}\log\frac{100}{100-99.99}$

= $\frac{2.303}{k}\log\frac{100}{0.01}$

= $\frac{2.303}{k}\log 10^4$

= $\frac{4\times 2.303}{k}$

(49) Answer : (1)

Hint:

Rate = k[NH_3]° = k

Solution:

The decomposition of gaseous ammonia on a hot platinum surface is a zero order reaction at high pressure.

For zero order reaction

$t_{1/2} = \frac{A_0}{2k}$

(50) Answer : (1)

Solution:

Rate of reaction = $\frac{1}{3}\frac{\mathrm{d}\left[\mathrm{H_2}\right]}{\mathrm{dt}}$

$0.6 = \frac{1}{3}\frac{\mathrm{d}\left[\mathrm{H_2}\right]}{\mathrm{dt}}$

$\frac{\mathrm{d}[\mathrm{H_2}]}{\mathrm{dt}} = 1.8 \text{ mol } \mathrm{L}^{-1}\mathrm{s}^{-1}$

(51) Answer : (1)

Solution:

Slow step is rate determining step

r = K[A_3][A]

$\mathrm{K_C} = \frac{[\mathrm{A_2}][\mathrm{A}]}{[\mathrm{A_3}]}$

$\left[\mathrm{A}\right] = \frac{\mathrm{K_C}[\mathrm{A_3}]}{[\mathrm{A_2}]}$

r = K KC $\frac{[\mathrm{A_3}]}{[\mathrm{A_2}]}$[$A_3$]

r = $K^1[A_3]^2[A_2]^{-1}$

(52) Answer : (3)

Solution:

r = $k[A]^x [B]^y$

4r = $k[2A]^x [2B]^y$

r = $k[A]^x [2B]^y$

On solving these equation

x = 2, y = 0

Hence,

r = $k[A]^2 [B]^0$

(53) Answer : (3)

Hint:

Rate = $PZ_{AB}\, e^{-E_a/RT}$

Solution:

$e^{-E_a/RT}$ represents the fraction of molecules with energies equal to or greater than E_a.

(54) Answer : (4)

Solution:

$\log\frac{\mathrm{k_2}}{\mathrm{k_1}} = \frac{\mathrm{Ea}}{2.303\,\mathrm{R}}\left(\frac{1}{\mathrm{T_1}} - \frac{1}{\mathrm{T_2}}\right)$

$\log 2 = \frac{\mathrm{Ea}}{2.303\times 8.314}\left(\frac{1}{300} - \frac{1}{310}\right)$

E_a = 0.3 × 2.303 × 8.314 × 300 × 31 J

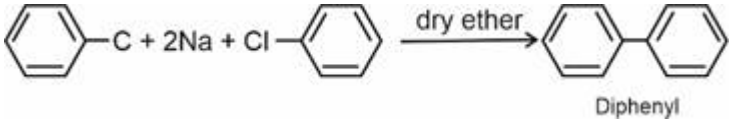
= 53.42 kJ

(55) Answer : (3)

Solution:

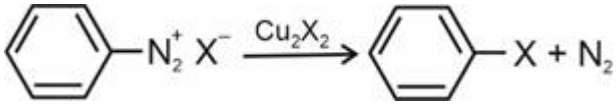
Unit of rate constant for nth order reaction is $(\text{mol L}^{-1})^{1-n}\ \text{s}^{-1}$

(56) Answer : (4)
Hint:
Finkelstein and Swarts reactions are halogen exchange.
Solution:
1.



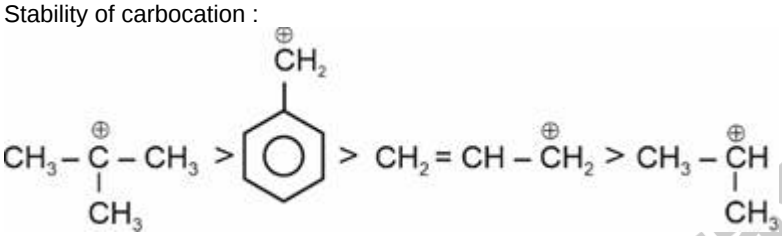
Fittig reaction
2. $\text{R–X} + \text{NaI} \rightarrow \text{R–I} + \text{Na}$

Finkelstein reaction
3. $\text{R–Br} + \text{Ag} + \text{F} \rightarrow \text{R–F} + \text{AgBr}$
Swarts reactions
4.



Sandmeyer reaction

(57) Answer : (4)
Hint:
Order to S_N1 depends upon stability of intermediate carbocation.
Solution:
Stability of carbocation :



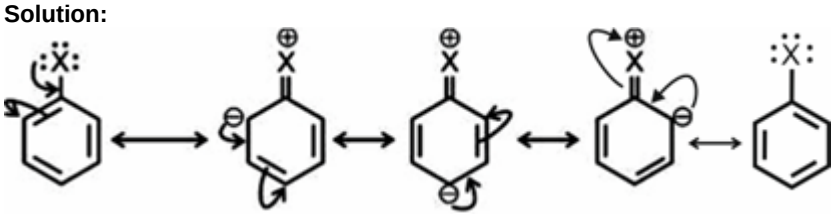
(58) Answer : (4)
Hint:
AgCN is a covalent compound and nitrogen is free to donate electron pair.
Solution:
In AgCN, nitrogen atom is free to donate electron pair forming isocyanide as the main product while reacting with alkyl halide
R – X + AgCN → RNC + Ag

(59) Answer : (2)
Solution:
Selectivity ratio for monochlorination with respect to 1°, 2°, 3° carbon is 1 : 3.8 : 5 respectively.
% primary halide = $\frac{9\times1}{9+5} \times 100 = 64.28\%$
% tertiary halide = $\frac{5}{14} \times 100 = 35.72\%$

(60) Answer : (4)
Solution:
S_N1 reactions proceeds via formation of carbocation intermediate.

(61) Answer : (2)
Solution:
With the increase in size and mass of halogen atom, the magnitude of Van der Waals forces increases, so methyl iodide has higher boiling point than methyl bromide.

(62) Answer : (3)
Hint:
In haloarenes, C–X bond acquires a partial double bond character due to resonance.
Solution:



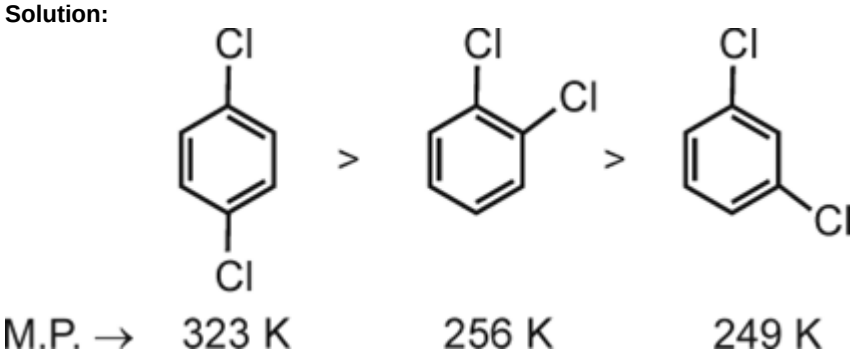
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(where X = halogen atom)
A bond cleavage in haloarenes is difficult than haloalkane and therefore, they are less reactive towards nucleophilic substitution reaction.

(63) **Answer :** (4)
Hint:
Oxidation of chloroform forms carbonyl chloride.
Solution:

$$2\,CHCl_3 + O_2 \xrightarrow{light} \underset{\text{Phosgene}}{2\,COCl_2} + 2\,HCl$$

(64) **Answer :** (3)
Solution:



(65) **Answer :** (4)
Solution:

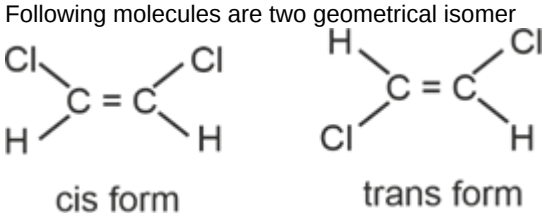
$$C_2H_5Br + 2Na + BrC_2H_5 \xrightarrow{\text{dry ether}} CH_3 - CH_2 - CH_2 - CH_3$$

(66) **Answer :** (2)
Solution:
Thin layer chromatography is a type of adsorption chromatography where as paper chromatography is type of partition chromatography.

(67) **Answer :** (4)
Hint:
According to IUPAC nomenclature carboxylic acid and its derivatives are at higher priority.
Solution:
Correct order of priority of functional groups in IUPAC naming is
– COOH > – COCl > – OH > – NH_2

(68) **Answer :** (2)
Hint:
Percentage of halogen (X)
= $\frac{atomic\ mass\ of\ X \times m_1 \times 100}{molecular\ mass\ of\ AgX \times m}$
m_1 = mass of AgX
m = mass of organic compound
Solution:
m_1 = 1.128 g, m = 1 g
M(AgBr) = 188 g mol^{-1}
% of Br = $\frac{80 \times 1.128 \times 100}{188 \times 1} = 48\%$

(69) **Answer :** (4)
Hint:
If the two atoms or groups attached to each sp^2 hybridised carbon atom of alkene are different, they can show geometrical isomerism.
Solution:
Following molecules are two geometrical isomer



(70) **Answer :** (2)
Hint:

$$\% \text{ of S} = \frac{32 \times \text{weight of } BaSO_4}{233 \times \text{weight of organic substance}} \times 100$$

Solution:

$\% \text{ of S} = \frac{32 \times 0.466}{233 \times 0.16} \times 100$

= 40%

(71) Answer : (3)

Hint:

Two or more compounds having same molecular formula but different functional groups are called functional isomers.

Solution:

A carboxylic acid and ester have same general formula but have different functional group.

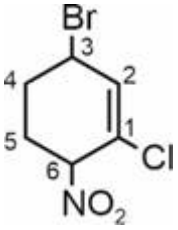
(72) Answer : (3)

(73) Answer : (3)

Hint:

Numbering is done in accordance with the lowest set of locant rule so that multiple bond should get lowest number.

Solution:



3-Bromo-1-chloro-6-nitrocyclohex-1-ene.

(74) Answer : (2)

Hint:

$$\% \text{ Nitrogen} = \frac{1.4 \times \text{No. of milliequivalent of acid}}{\text{wt. of organic compound}}$$

Solution:

$\% \text{ Nitrogen} = \frac{1.4 \times 20 \times 0.6 \times 2}{0.8}$

$= \frac{1.4 \times 20 \times 1.2}{0.8}$

$= \frac{1.4 \times 20 \times 12}{8}$

= 14 × 3 = 42%

(75) Answer : (3)

Hint:

Tropylium cation is an aromatic species.

Solution:



, cyclopentadienyl cation is an antiaromatic species.

(76) Answer : (4)

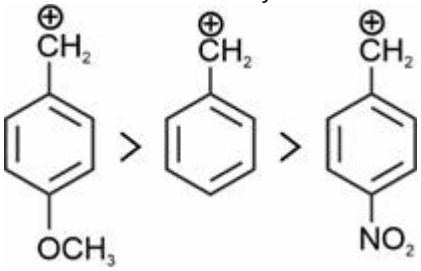
Hint:

Electron donating group present at para position, in the benzene moiety, stabilises the carbocation

Solution:

• $-OCH_3$ is electron donating group while $-NO_2$ is electron withdrawing group.

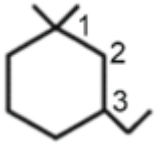
• Correct order of stability of carbocations:



(77) Answer : (1)

Solution:

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3-Ethyl-1, 1-dimethylcyclohexane.

(78) **Answer :** (3)

Hint:

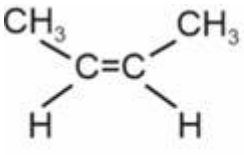
Nucleophiles are the electron-rich species which have either negative charge or lone pair of electrons.

Solution:

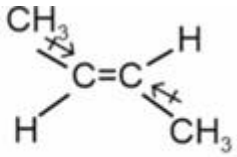
$AlCl_3$ is an electron-deficient species which acts as electrophile.

(79) **Answer :** (3)

Hint:

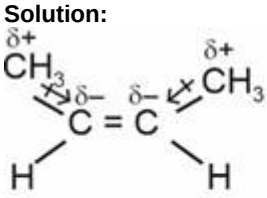


cis-But-2-ene

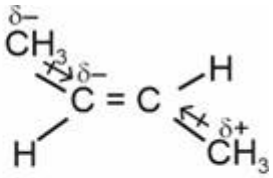


trans-But-2-ene

Solution:



cis-But-2-ene
(μ = 0.33D)
B.P = 277 K



trans-But-2-ene
(μ = 0)
B.P = 274 K

Due to more symmetry, trans isomer has higher melting point in solid state.

(80) **Answer :** (3)

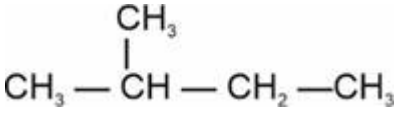
Hint:

The boiling point of straight chain alkanes increases with increase in carbon atoms while boiling point of isomeric alkanes, decreases with increasing branching due to less van der Waals forces.

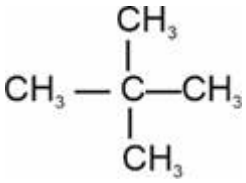
Solution:

$H_3C–CH_2–CH_2–CH_2–CH_2–CH_3$ 341.9 K
n-Hexane

$CH_3–CH_2–CH_2–CH_2–CH_3$ 309.1 K
n-pentane



300.9 K
2-methylbutane



282.5 K
2, 2-dimethylpropane

(81) **Answer :** (1)

Solution:

Magnitude of torsional strain depends upon the angle of rotation about C – C bond. This angle is also called dihedral angle or torsional angle.

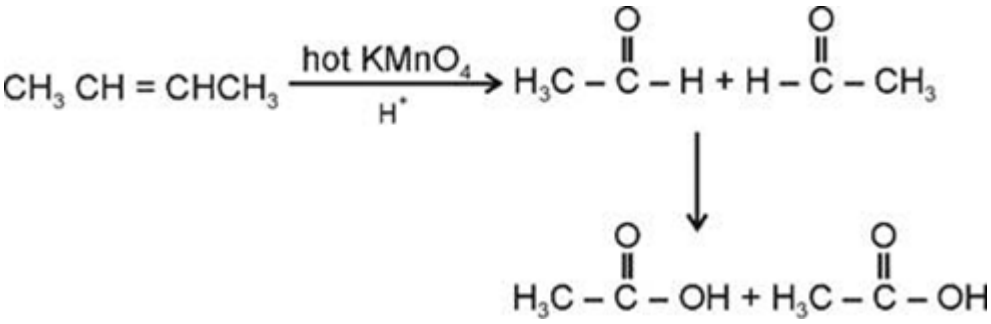
(82) **Answer :** (2)

Hint:

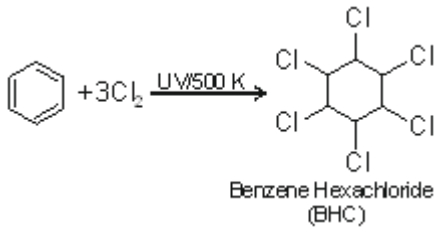
Aldehydes are further oxidised to acids in presence of acidified $KMnO_4$.

Solution:

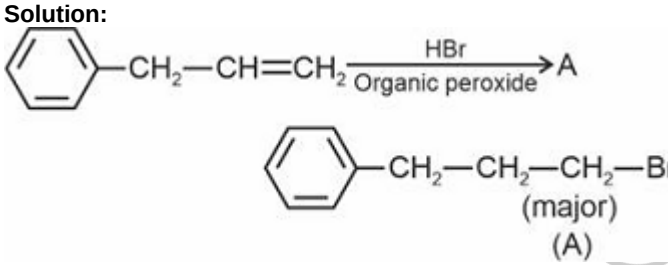
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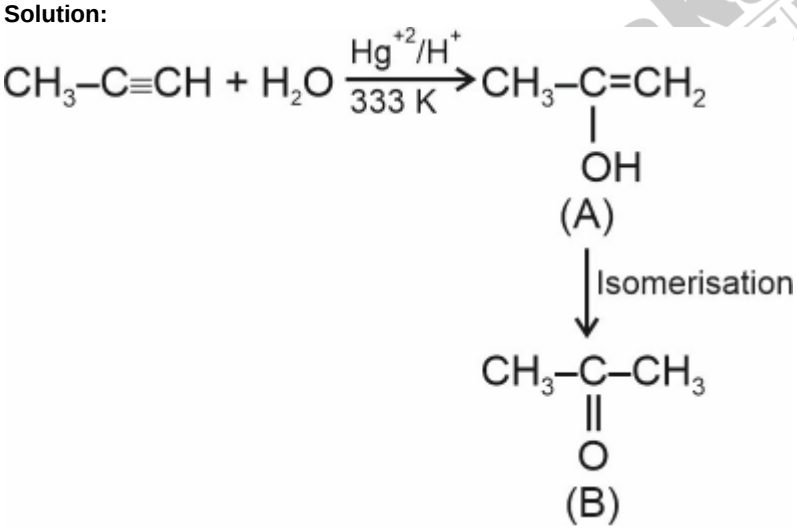
(83) Answer : (3)
Solution:



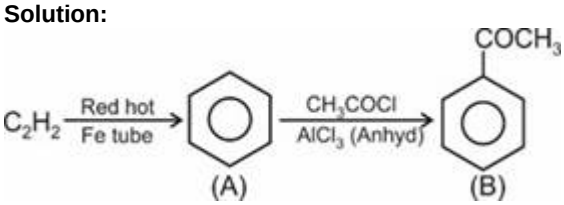
(84) Answer : (3)
Hint:
In the presence of peroxide, addition of HBr to unsymmetrical alkenes follows the anti Markovnikov rule.
Solution:



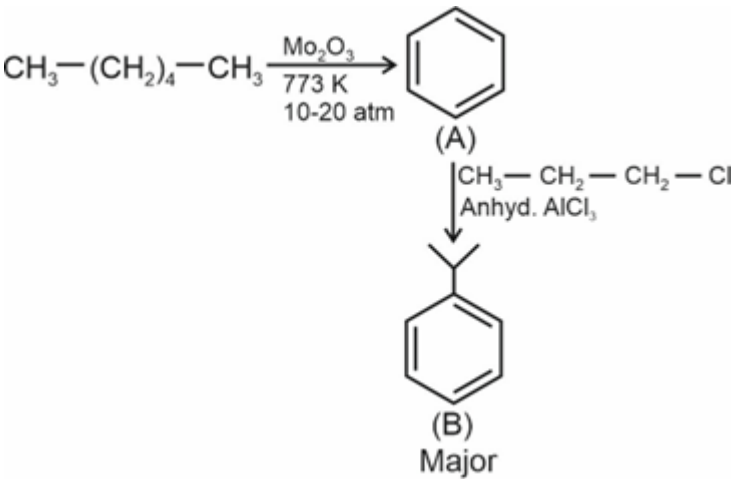
(85) Answer : (4)
Solution:



(86) Answer : (3)
Solution:



(87) Answer : (4)
Solution:

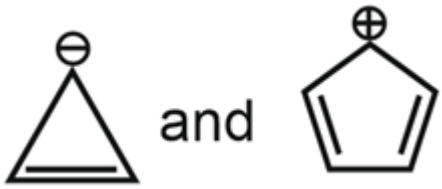


(88) **Answer :** (2)
Solution:

$$2\,CH_4 + O_2 \xrightarrow{Cu\ /523K/100\ atm} 2\,CH_3\,OH$$

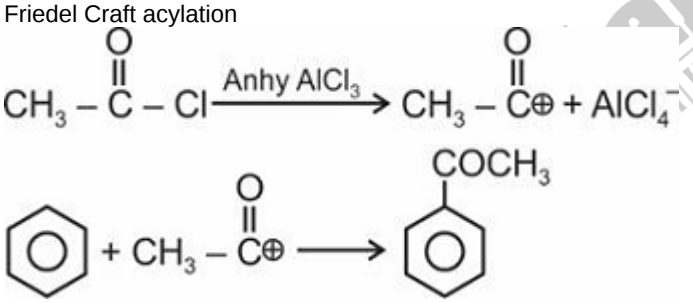
$$CH_4 + O_2 \xrightarrow[\Delta]{Mo_2\ O_3} \underset{(B)}{HCHO} + H_2O$$

(89) **Answer :** (3)
Solution:



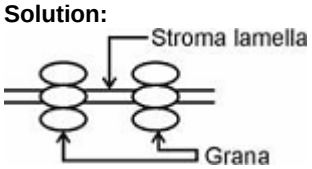
are antiaromatic species.

(90) **Answer :** (4)
Solution:
Friedel Craft acylation



BOTANY

(91) **Answer :** (1)
Solution:



(92) **Answer :** (4)
Hint:
The ATP synthase enzyme consists of two parts: CF_0 and CF_1.
CF_0 forms transmembrane channel.
CF_1 protrudes from the outer surface of thylakoid membrane (stroma side).
Solution:
A conformational change in the CF_1 particle of the ATP synthase, results in synthesis of ATP molecules.

(93) **Answer :** (3)
Solution:
Joseph Priestley conducted the bell jar experiment to prove that plants restore to the air whatever breathing animals and burning candle remove.

(94) **Answer :** (4)
Hint:
C_3 plants show photorespiration.
Solution:
In C_4 plants, photorespiration does not occur. This is because these plants have a mechanism that increases the concentration of CO_2 at the enzyme site. During the C_4 pathway, when the C_4 acid from the mesophyll cells is broken down in the bundle sheath cells, it releases CO_2 – this results in increasing the intracellular concentration of CO_2. This in turn, ensures that the RuBisCO functions as a carboxylase minimising the oxygenase activity. Thus, the productivity and yields are better in C_4 plants as compared to C_3 plants.

(95) **Answer :** (3)
Solution:
In Z-scheme (non-cyclic photophosphorylation)
• Both PS-I and PS-II are involved
• Splitting of water is associated with PS-II
• Electrons in Z scheme of light reaction move both uphill and downhill on redox potential scale.

(96) **Answer :** (2)
Solution:
Chlorophyll *a* is the primary photosynthetic pigment. Chlorophyll *b* accounts for about one fourth of total chlorophyll content and appears yellow-green in chromatogram. Xanthophyll appears yellow in the chromatogram. Carotenoids protect the plant from excessive heat and prevent photo-oxidation of reaction centre.

(97) **Answer :** (1)
Solution:
In C_4-plants, photorespiration does not occur. During photorespiration, there is neither synthesis of sugar and ATP nor NADPH.

(98) **Answer :** (4)
Solution:
The diagram is of C_4 pathway, in which P represents fixation, Q represents decarboxylation and R represents the regeneration process.

(99) **Answer :** (4)
Solution:
Light reactions are directly light driven and dark reactions are not directly light driven but are dependent on the products of light reaction. Dark reaction occurs in the stroma while light reaction occur in the membrane of thylakoids.

(100) **Answer :** (3)
Solution:
ATP synthesis in chloroplast takes place towards the stroma side of the thylakoid membrane.

(101) **Answer :** (1)
Solution:
Accessory pigments are all pigments except chlorophyll *a*, they absorb light of different wavelength and protect chlorophyll *a* from photo-oxidative damage.

(102) **Answer :** (3)
Solution:
Tomato is a C_3 plant. In C_3 plant, formation of one glucose molecule requires 18 ATP and 12 $NADPH_2$.
$\therefore$ 90 ATP and 60 NADPH are required for the formation of five molecules of glucose.

(103) **Answer :** (2)
Hint:
Formation of acetyl CoA occurs in the matrix of mitochondria.
Solution:
The complete oxidation of pyruvate by the stepwise removal of all the hydrogen atoms, leaving three molecules of CO_2 occurs in 2 steps:
(i) Formation of Acetyl coenzyme A
(ii) TCA cycle
This process takes place in the matrix of mitochondria.

(104) **Answer :** (1)
Hint:
The reduced ubiquinone is called ubiquinol.
Solution:

The reduced ubiquinone (Ubiquinol) is oxidised with the transfer of electrons to cytochrome *c* via cytochrome bc_1 complex (complex III).

(105) Answer : (2)

Hint:

1 glucose $\xrightarrow{\text{Zymase}}$ 2 alcohol + 2 CO_2

Solution:

Aerobic respiration

$C_6H_{12}O_6 + 6\ O_2 \rightarrow 6\ CO_2 + 6\ H_2O$

(106) Answer : (2)

Hint:

CO_2 is not released during lactic acid fermentation.

Solution:

In both lactic acid and alcohol fermentation not much energy is released ; less than seven percent of the energy in glucose is released and not all of it is trapped as high energy bonds of ATP. Net gain in both the types of fermentation is 2ATP.

(107) Answer : (3)

Solution:

During conversion of oxalosuccinic acid to α-KGA and α-KGA to Succinyl CoA, CO_2 is evolved

Pyruvic acid → Acetyl CoA is link reaction.

(108) Answer : (2)

Hint:

Fermentation accounts for only partial breakdown of glucose.

Solution:

Pyruvate, after it enters mitochondrial matrix undergoes oxidative decarboxylation by a complex set of reactions catalyzed by pyruvic dehydrogenase. Conversion of succinyl CoA to succinic acid is an example of substrate level phosphorylation. The role of oxygen is limited to the terminal stage of the process. Yet, the presence of oxygen is vital, since it drives the whole process by removing hydrogen from the system. Oxygen acts as the final hydrogen acceptor.

(109) Answer : (1)

Solution:

The first member of citric acid cycle is oxaloacetic acid.

(110) Answer : (4)

Solution:

When 3-phosphoglyceraldehyde is converted to 1, 3-bisphosphoglycerate, two redox-equivalents are removed (in the form of two hydrogen atoms) and thus NADH + H^+ is formed.

(111) Answer : (3)

Hint:

Proteins are degraded into amino acids and then can connect to TCA as pyruvate or as acetyl CoA

Solution:

Fat → fatty acid + glycerol

Fatty acid → acetyl CoA

Both fat and protein can combine to a common intermediate, acetyl CoA and then join TCA cycle.

(112) Answer : (4)

Hint:

Cytochrome *c* is mobile electron carrier.

Solution:

Cytochrome *c* is a small protein attached to the outer surface of the inner membrane of mitochondria and acts as a mobile carrier for transfer of electrons between complex III & IV.

(113) Answer : (3)

Hint:

Degraded product of autoclaved herring sperm DNA had a powerful cytokinesis promoting effect.

Solution:

Miller *et al.* discovered the first cytokinin from degraded product of autoclaved herring sperm DNA.

(114) Answer : (4)

Hint:

Apical bud secretes auxin which inhibits the growth of lateral bud.

Solution:

If apical bud is removed then auxin synthesis is inhibited due to which apical dominance is prevented and the lateral buds will sprout, *i.e.*, start growing.

(115) Answer : (1)

Hint:

The given diagram shows arithmetic growth.

Solution:
Arithmetic growth is not shown by all cells, tissues and organ of the plant.

(116) Answer : (2)
Hint:
Gibberellins are useful for quicker maturity in juvenile conifers.
Solution:
In juvenile conifers, gibberellins have been found to cause quicker early growth so that maturity is reached early.

(117) Answer : (3)
Hint:
Growth of the given system per unit time expressed on a common basis, *e.g.*, per unit initial parameter is called the relative growth rate.
Solution:
In leaf A
Initial size = 10 cm^2
Growth per unit time = 5 cm^2
Relative Growth Rate (RGR) = $\frac{\text{Growth per unit time}}{\text{Initial size}} \times 100$
RGR of leaf A = $\frac{5}{10} \times 100 = 50\%$
In leaf B
Initial size = 25 cm^2
Growth per unit time = 5 cm^2
RGR of leaf B = $\frac{5}{25} \times 100 = 20\%$

(118) Answer : (2)
Hint:
ABA acts as a general plant growth inhibitor.
Solution:
Auxin promotes flowering in pineapples. Ethylene is used to initiate flowering and for synchronising fruit set in pineapples.

(119) Answer : (1)
Solution:
Aqueous solution of ethephon is readily absorbed by plants and releases ethylene slowly.

(120) Answer : (3)
Solution:
ABA is also known as stress hormone as it increases the tolerance of plants in adverse environmental conditions

(121) Answer : (1)
Solution:
Buttercup exhibits heterophyllous development due to environment.

(122) Answer : (3)
Solution:
In the geometric growth equation.
W_1 = Final size, W_0 = Initial size, r = Growth rate, t = Time of growth and e = Base of natural logarithm.

(123) Answer : (2)
Hint:
Auxins do not have metabolic role inside the human body.
Solution:
Auxin was first isolated from human urine samples because auxins do not have metabolic role so excreted out of the body.

(124) Answer : (1)
Solution:
As an outbreeding device, self incompatibility is genetically controlled mechanism.

(125) Answer : (2)
Hint:
Hilum represents junction between ovule and funicle.
Solution:
The body of the ovule fuses with funicle in the region called hilum.

(126) Answer : (1)
Hint:
Seed dormancy allows seeds to overcome periods that are unfavourable for seedling establishment.
Solution:
Seeds form the basis of our agriculture as they show dehydration and dormancy that helps in storage of seeds which can be used as food throughout the year and also to raise crops in the next season.

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(127) Answer : (3)

Hint:

In most of angiosperms, the pollens are shed at 2-celled stage.

Solution:

Asymmetric mitosis in pollen results in vegetative and generative cell *i.e.*, 2-celled stage. Pollination at 2-celled stage occurs in 60% of angiosperms.

(128) Answer : (3)

Solution:

Apomixis is a form of asexual reproduction. So, there is no segregation of characters.

(129) Answer : (4)

Hint:

Vallisneria shows epihydrophily.

Solution:

In *Vallisneria*, light and unwettable pollen grains are present. Male flowers, after breakage float on the surface of water and pollens are carried passively by water currents. Hence, pollination takes place on the surface of water.

(130) Answer : (3)

Hint:

Megaspore mother cell is diploid

Solution:

The correct sequence of embryo sac formation is

Megasporangium (2n) → Megaspore mother cell (2n) $\xrightarrow{meiosis}$ Megaspore (n) → Embryo sac (n)

(131) Answer : (4)

Hint:

Michelia shows apocarpous condition.

Solution:

In *Michelia*, gynoecium is multicarpellary and carpels are free from each other.

(132) Answer : (4)

Hint:

The two anther lobes are separated by a deep grove in front and are attached to each other by a band of vesiculated sterile tissue called connective.

Solution:

The proximal end of the filament is attached to the thalamus or the petal of the flower.

(133) Answer : (4)

Solution:

Water hyacinth and water lily are insect pollinated flowers.

Hydrilla plant is water pollinated while grasses are pollinated by wind.

(134) Answer : (2)

Solution:

Vegetative cell of pollen grain is larger in size than generative cell and has irregularly shaped nucleus.

(135) Answer : (3)

Solution:

In seeds of castor, the endosperm persist in the mature seeds, thus, they are known as albuminous seed.

ZOOLOGY

(136) Answer : (3)

Solution:

Pennatula, *Obelia*, *Meandrina* and *Aurelia* are coelenterates.

They exhibit tissue level of body organisation.

Aurelia has a medusa form.

Metagenesis is seen in *Obelia.* Polyps produce medusae asexually and medusae form the polyps sexually.

(137) Answer : (2)

Solution:

Porifers are acoelomates but they are mostly asymmetrical. Coelenterates and ctenophores are radially symmetrical but they exhibit tissue level of body organization.

(138) Answer : (2)

Solution:

Water vascular system is present in members of phylum Echinodermata.

In echinoderms, digestive system is complete whereas excretory system is absent.

Adults are radially symmetrical whereas larval forms are bilaterally symmetrical.

They are dioecious and have sexual mode of reproduction.
Development is indirect with free swimming larva.

(139) Answer : (2)
Solution:
The name cnidaria is derived from the cnidoblasts or cnidocytes (which contain the stinging capsules or nematocysts) present on the tentacles and the body of cnidarians. Cnidoblasts are used for anchorage, defence and for the capture of prey.

(140) Answer : (4)
Solution:
Meandrina (Brain coral) is a coelenterate/cnidarian. It has a central gastrovascular cavity with a single opening, mouth on hypostome.
Cuttle fish (Mollusc) is dioecious.
The body of a scorpion (Arthropod) is divided into cephalothorax and abdomen.
Hooks are not present in liver fluke (Platyhelminth).

(141) Answer : (2)
Solution:
The given statement talks about the features of molluscs. *e.g.-Pila* and *Dentalium*.

(142) Answer : (3)
Solution:
Collar cells or choanocytes line the spongocoel and the water canals. Metagenesis or alternation of generation is exhibited by some cnidarians *e.g.*, *Obelia*. Ctenophores are commonly known as sea walnuts or comb jellies.

(143) Answer : (2)
Solution:
The adult echinoderms are radially symmetrical but their larvae are bilaterally symmetrical. *Aplysia* is a mollusc and *Anopheles* is an arthropod.

(144) Answer : (2)
Solution:
Ascaris, *Pheretima* and *Pila* exhibit organ-system level of body organization.
Platyhelminthes- Organ level of body organisation
Ctenophores/cnidarians- Tissue level of body organisation
Porifera- Cellular level of body organisation

(145) Answer : (2)
Solution:
The body of *Ancylostoma* (Aschelminth) is circular in cross-section and possesses pseudocoelom. They have a complete alimentary canal and organ-system level of body organisation. It infects intestine of humans.

(146) Answer : (2)
Solution:
In echinoderms, digestive system is complete with mouth on the lower (ventral) side and anus on the upper (dorsal) side.

(147) Answer : (3)
Hint:
Annelida
Solution:
Nereis belongs to the phylum Annelida. They may be aquatic or terrestrial, free-living, and sometimes parasitic. They exhibit organ-system level of body organisation and bilateral symmetry. They are triploblastic, metamerically segmented and coelomate animals.

(148) Answer : (3)
Solution:
Hirudinaria (Blood sucking leech) is monoecious in nature and performs sexual reproduction.

(149) Answer : (3)
Solution:
Sponges undergo internal fertilisation. The development in sponges is mediated by an intermediate free-swimming flagellated larval stage for dispersal of species. Thus, the development in sponges is indirect. Their larvae are morphologically distinct from adults. Both asexual and sexual reproduction occur in sponges.

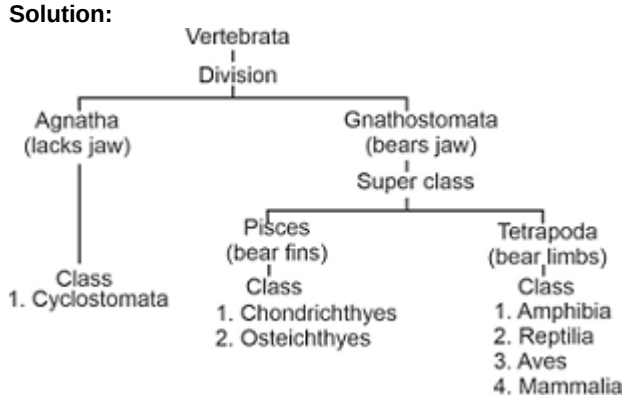
(150) Answer : (3)
Solution:
Both echinoderms (*Echinus*)and hemichordates (*Balanoglossus*) are exclusively marine.

(151) Answer : (3)
Solution:
Both (A) and (R) are true and (R) is the correct explanation of (A).
The members of the subphylum vertebrata possess notochord during the embryonic period. The notochord is replaced by a cartilaginous or bony vertebral column in the adults.
Thus, all vertebrates are chordates but all chordates are not vertebrates.

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(152) Answer : (1)

Solution:



(153) Answer : (3)

Hint:

Rana belongs to this class.

Solution:

Amphibians have the following features:-
(i) Body is divisible into head and trunk.
(ii) A tympanum represents the ear.
(iii) Respiration is by gills, lungs and through skin.
(iv) Sexes are separate.
(v) Oviparity is seen
(vi) Skin is moist without scales

(154) Answer : (2)

Hint:

It is a reptile.

Solution:

Generally reptiles have a three-chambered heart and show incomplete double circulation.
Crocodilus is an exception. It has 4-chambered heart.

(155) Answer : (1)

Hint:

Helps fishes to swim

Solution:

Members of the class Chondrichthyes as well as Osteichthyes have streamlined body. Chondrichthyes (males) have claspers in the pelvic fins. Their skin is tough, containing minute placoid scales.
Operculum (gill cover) is present in the members of the class Osteichthyes.

(156) Answer : (3)

Hint:

Birds and mammals are warm-blooded animals.

Solution:

Ichthyophis, *Bufo* and *Rana* belong to the class Amphibia while *Calotes* and *Chameleon* belong to the class Reptilia. Amphibians and reptiles are poikilotherms.
Aptenodytes belongs to the class Aves while *Pteropus* and *Macropus* belong to the class Mammalia. Birds and mammals are homeothermous.

(157) Answer : (2)

Hint:

Function of modified forelimbs.

Solution:

The forelimbs of birds are modified into wings. The hind limbs generally have scales and are modified for walking, swimming and clasping the tree branches.

(158) Answer : (3)

Hint:

Internal fertilisation occurs in *Pristis.*

Solution:

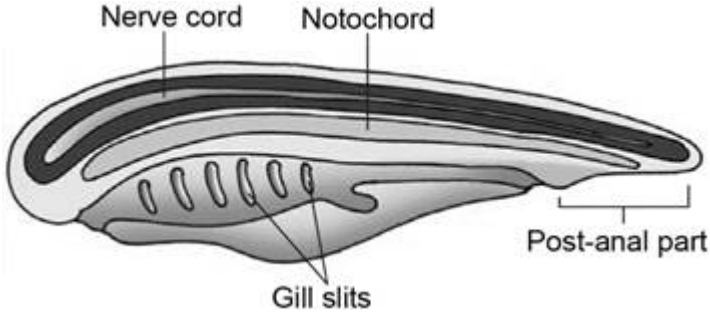
Bony fishes	**Cartilaginous fishes**
Mouth is mostly terminal	Mouth is located ventrally
Skin is covered with cycloid/ctenoid scales	Skin is tough, containing minute placoid scales
Fertilisation is usually external	Fertilisation is usually internal
Claspers are absent	Male pelvic fins bear claspers

(159) Answer : (3)

Hint:

Nerve cord is dorsally located

Solution:



In members of the phylum Chordata, notochord is ventral to nerve cord and dorsal to gut.

(160) Answer : (1)

Solution:

• Frogs have webbed digits that help in swimming.
• Body of a frog is divisible into head and trunk. A neck and tail are absent.
• On either side of eyes, a membranous tympanum (ear) receives sound signals.

(161) Answer : (3)

Solution:

Internal fertilisation and oviparity is exhibited by members belonging to the class Aves. Presence of hair on skin is a unique mammalian feature. *Ornithorhynchus* exhibits oviparity. *Aptenodytes* and *Neophron* (birds) do not possess hair on their skin.

(162) Answer : (4)

Solution:

The members of order Crocodilia as well as class Aves and Mammalia possess four-chambered heart. Rest of the members of the class Reptilia have three-chambered heart.

(163) Answer : (4)

Solution:

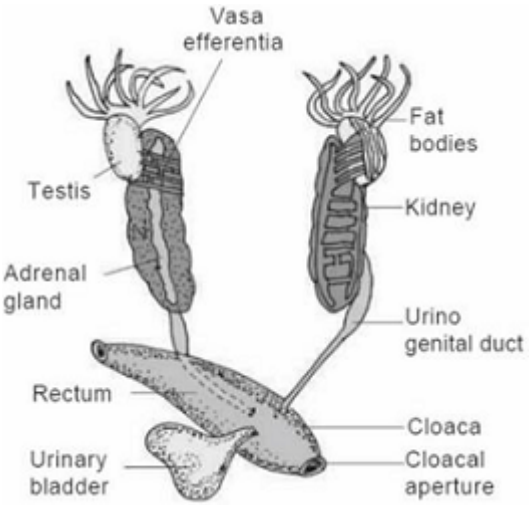
Animals belonging to class Cyclostomata are ectoparasites on some fishes. They are marine but migrate to fresh water for spawning. Their larvae after metamorphosis return to the ocean. Cyclostomes are vertebrates.

(164) Answer : (3)

Hint:

Vasa efferentia open into Bidder's canal.

Solution:



(165) Answer : (3)

Hint:

Also seen in reptiles, birds and mammals.

Solution:

Frogs perform pulmonary, cutaneous and buccal respiration when they are present on land but perform only cutaneous respiration when present in water, during aestivation and hibernation.

(166) Answer : (1)

Hint:

Absent in birds and mammals

Solution:

Special venous connection between liver and intestine as well as the kidney and lower parts of the body are present in frogs. The former is called hepatic portal system and the latter is called renal portal system. In humans, hepatic portal

system is present but renal portal system is absent.

(167) Answer : (2)

Hint:

Cutaneous respiration

Solution:

Skin of frog is moist due to the presence of mucus glands and is highly vascular. So, it participates in gaseous exchange in water as well as on land. Gills are absent in adult frogs. Lungs are also highly vascular structures but frog performs pulmonary respiration when it is present on land in normal climatic conditions.

(168) Answer : (2)

Hint:

Rana tigrina has a bony endoskeleton.

Solution:

Hind brain consists of cerebellum and medulla oblongata. There are 10 pairs of cranial nerves arising from the brain of frogs.

Brain is enclosed in a bony structure called the brain box.

Pons is absent in frogs.

(169) Answer : (1)

Solution:

Frog has different types of sense organs, namely organs of touch (sensory papillae), taste (taste buds), smell (nasal epithelium), vision (eyes) and hearing (tympanum with internal ears). Out of these eyes and internal ears are well organised structures and the rest are cellular aggregations around nerve endings.

(170) Answer : (3)

Hint:

Limbs without webs in between digits

Solution:

In frogs, the forelimbs and hind limbs help in swimming, walking, leaping and burrowing. Forelimbs have four digits while hind limbs have five digits in both male and female frogs. Frogs exhibit sexual dimorphism. Male frogs can be distinguished by the presence of sound producing vocal sacs and also a copulatory pad on the first digit of the fore limbs which are absent in female frogs.

(171) Answer : (1)

Hint:

Herbivores have long intestine.

Solution:

The digestive system of frogs consists of alimentary canal and digestive glands. The alimentary canal is short because frogs are carnivores and hence the length of their intestine gets reduced as compared to a tadpole.

(172) Answer : (3)

Hint:

Recall the position of heart in chordates

Solution:

In the excretory system of a frog, the thin-walled urinary bladder is present ventral to the rectum which also opens in the cloaca.

(173) Answer : (1)

Hint:

Largest phallomere of male cockroach.

Solution:

Pseudopenis and titillator are parts of left phallomere which is considered as the largest phallomere of male cockroach.

(174) Answer : (4)

Hint:

Gizzard is distal to crop from mouth.

Solution:

In cockroach, oesophagus opens into sac-like structure called crop which is followed by gizzard or proventriculus. Gizzard helps in grinding the food particles. Gizzard has an outer layer of thick circular muscles and thick inner cuticle forming six highly chitinous plates called teeth.

(175) Answer : (1)

Hint:

Segment in which spermatheca is present in females

Solution:

Testes – Present on each lateral side in the 4th-6th abdominal segments.

Anal cerci – Present in the 10th segment of both sexes.

Mushroom gland – Present in the 6th-7th abdominal segments in males

Ovaries – Present in the 2nd-6th abdominal segments of females

(176) Answer : (4)

Hint:

Only adult cockroach has fully developed wings

Solution:

On an average, a female cockroach produces 9-10 oothecae, each containing 14-16 eggs. The development of *P.americana* is paurometabolous, meaning there is development through nymphal stages. The nymphs look very much like adults. The nymph grows by moulting about 13 times to reach the adult form. The next to last nymphal stage has wing pads but only adult cockroaches have wings.

(177) Answer : (2)

Hint:

Main excretory product of frog is same as in humans.

Solution:

Cockroach is uricotelic whereas frog is ureotelic.

Malpighian tubules present in cockroach absorb nitrogenous waste products and convert them into uric acid which is excreted out through the hindgut. Therefore, this insect is uricotelic.

The frog excretes urea and is therefore, ureotelic.

(178) Answer : (3)

Solution:

Nervous system lies mid ventrally. Mosaic vision offers more sensitivity but less resolution.

(179) Answer : (2)

Hint:

Number is 2

Solution:

Mandibles possess grinding and incising regions. Cockroaches possess a pair of mandibles and a pair of maxillae.

Ostia are the apertures present at the end of each heart chamber. Proventriculus is another name for gizzard.

(180) Answer : (1)

Solution:

The 7th sternum of female cockroach is boat-shaped.

In male cockroaches, the mushroom-shaped gland is present in the 6th-7th abdominal segments and a pair of testes lie in the 4th-6th abdominal segments.

In female cockroaches, the ovaries lie in the 2nd-6th abdominal segments.

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